

Recursion - Array Question

Q> Search array is sorted or not.

arr [1, 2, 3, 4, 8, 9, 12]

1 < 2 $\neq$ arr [2, 3, 4, 8, 9, 12]

2 < 3 $\neq$ arr [3, 4, 8, 9, 12]

3 < 4 $\neq$ arr [4, 8, 9, 12]

4 < 8 $\neq$ arr [8, 9, 12]

8 < 9 $\neq$ arr [9, 12]

9 < 12 $\neq$ 0 return true

if arr length - 1 = i then return true

Q> Linear Search.

arr [1, 2, 3, 4] target = 4

arr[0] != 4, arr[2, 3, 4] - search in remaining array

arr[1] != 4, arr [3, 4]

arr[2] != 4, arr[4]

arr[3] == 4, return index.

Base Case

arr.length == i return -1;

- Rotated Binary Search.

arr = $\begin{bmatrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 6 & 7 & 8 & 9 & 1 & 2 & 3 \end{bmatrix}$

target = 7

s = 0 e = 7

mid = 3.

target != arr[mid]

(i) if arr[s] ≤ arr[mid]

if target ≥ arr[s] && target ≤ arr[mid]

end = mid - 1

else

s = mid + 1

(ii) ~~if arr[s] > arr[mid]~~

else

if (target > arr[mid] && target ≤ arr[end]) {

s = mid + 1

else

end = mid - 1;

- Linear Search. - Search from last;

• $arr = [1, 2, 3, 4, 4, 5]$

function (arr, target, index)

index = arr.length;

if (arr[index] == target) return index;

• search (arr, target, --index)

base condition

(i < 0) return -1;

- ~~Linear Search Occurrence~~

- Linear Search :- All Occurrence using ArrayList

⇒ $arr = [1, 2, 3, 4, 4, 4, 5];$

function (arr, ArrayList<Integer> list, target, i)

~~if~~ if (arr[i] == target) add i into list

~~return~~

~~if~~ then pass this list to function. with i+1;

function (arr, list, target, ++i);

Base Condition.

(i == arr.length) return list;

- All occurrence in Arry without using ArrayList in parameters.

```
function (arr, target, i)
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```
    ArrayList<Integer> list = new ArrayList<>();
```

```
    if (arr[i] == target) list.add(i);
```

```
    now
```

```
    add all occurrence of target from previous  
    next call.
```

```
    return list.
```

Base case :

```
if (index(i) == arr.length) return new ArrayList<>();
```